Magnetism in DFT: Three Key Challenges — One (Hidden) Exact Condition

Stefano Pittalis
CNR-NANO S3 Modena, Italy

- **Challenge 1:**
  Over-magnetization of MGGAs

- **Challenge 2:**
  Spurious xc-torques in Spin-DFT

- **Challenge 3:**
  Spin-orbit-coupling is not (always) “small”

Team

K. Bencheikh

G. Ambrogio

J. Maul

B. Civalleri

A. Erba

A. Boccuni

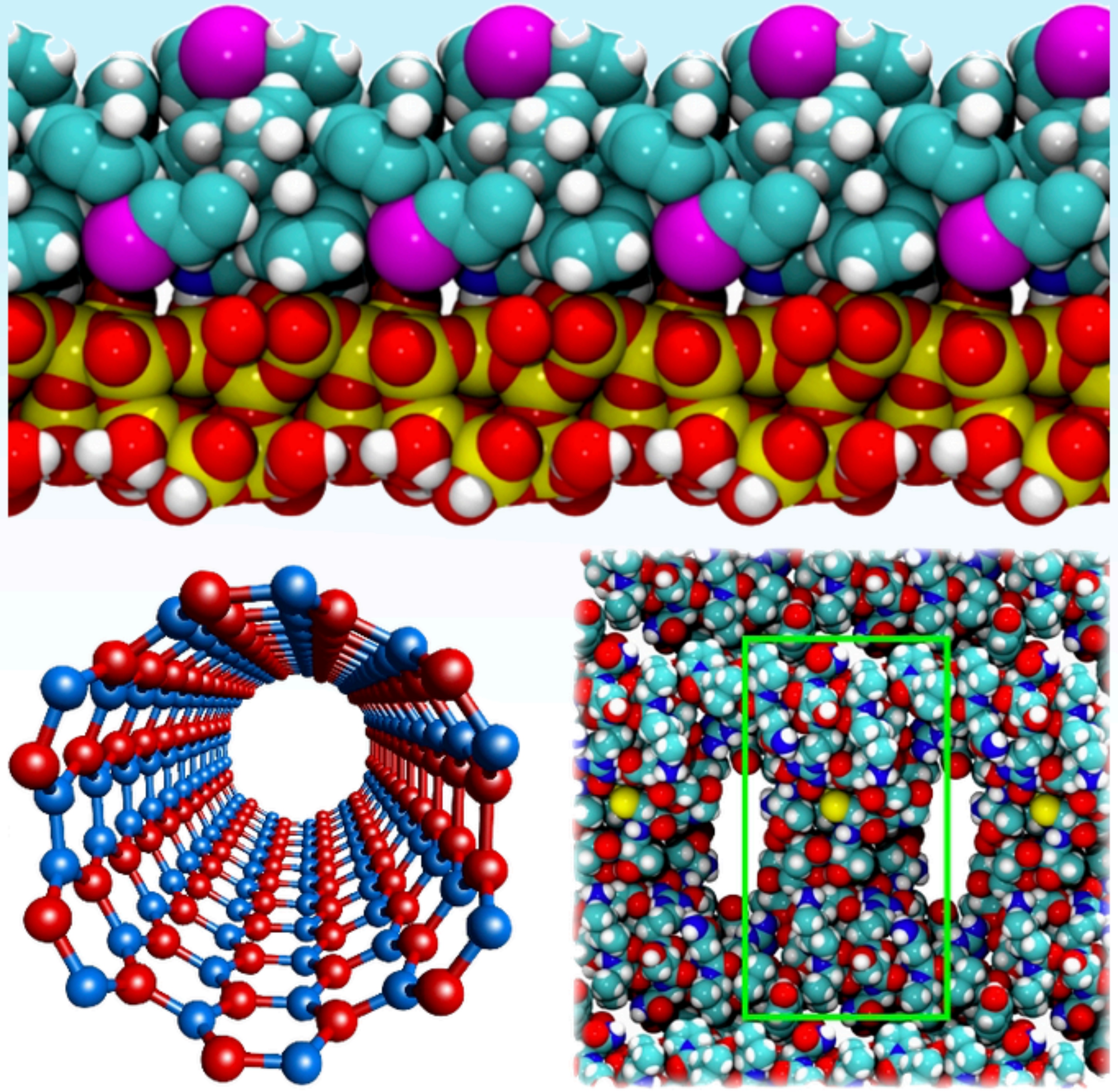
F.G. Eich

J.K. Desmarais

G. Vignale

F. Bodo

Code



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PHYSICAL REVIEW B **96**, 035141 (2017)

U(1)×SU(2) gauge invariance made simple for density functional approximations

S. Pittalis,[1,*] G. Vignale,[2] and F. G. Eich[3]
[1]*Istituto Nanoscienze, Consiglio Nazionale delle Ricerche, Via Campi 213A, I-41125 Modena, Italy*
[2]*Department of Physics and Astronomy, University of Missouri, Columbia, Missouri 65211, USA*
[3]*Max Planck Institute for the Structure and Dynamics of Matter, Luruper Chaussee 149, D-22761 Hamburg, Germany*
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PHYSICAL REVIEW MATERIALS **8**, 013802 (2024)

Generalized Kohn-Sham approach for the electronic band structure of spin-orbit coupled materials

Jacques K. Desmarais,[1,*] Giacomo Ambrogio,[1] Giovanni Vignale,[2] Alessandro Erba,[1] and Stefano Pittalis[3,†]
[1]*Dipartimento di Chimica, Università di Torino, via Giuria 5, 10125 Torino, Italy*
[2]*Institute for Functional Intelligent Materials, National University of Singapore, 4 Science Drive 2, 117544 Singapore*
[3]*Istituto Nanoscienze, Consiglio Nazionale delle Ricerche, Via Campi 213A, I-41125 Modena, Italy*
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PHYSICAL REVIEW LETTERS **132**, 256401 (2024)

Spin Currents via the Gauge Principle for Meta-Generalized Gradient Exchange-Correlation Functionals

Jacques K. Desmarais,[1,*] Jefferson Maul,[1] Bartolomeo Civalleri,[1] Alessandro Erba,[1] Giovanni Vignale,[2] and Stefano Pittalis[3,†]
[1]*Dipartimento di Chimica, Università di Torino, via Giuria 5, 10125 Torino, Italy*
[2]*Institute for Functional Intelligent Materials, National University of Singapore, 4 Science Drive 2, Singapore 117544*
[3]*Istituto Nanoscienze, Consiglio Nazionale delle Ricerche, Via Campi 213A, I-41125 Modena, Italy*
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Electron Localization Function for Noncollinear Spins

Jacques K. Desmarais,[1,*] Giovanni Vignale,[2] Kamel Bencheikh,[3] Alessandro Erba,[1] and Stefano Pittalis[4,†]
[1]*Dipartimento di Chimica, Università di Torino, via Giuria 5, 10125 Torino, Italy*
[2]*Institute for Functional Intelligent Materials, National University of Singapore, 4 Science Drive 2, Singapore 117544, Singapore*
[3]*Setif 1 University-Ferhat Abbas, Faculty of Sciences, Department of Physics and Laboratory of Quantum Physics and Dynamical Systems, Setif, Algeria*
[4]*Istituto Nanoscienze, Consiglio Nazionale delle Ricerche, Via Campi 213A, I-41125 Modena, Italy*
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PHYSICAL REVIEW LETTERS **134**, 106402 (2025)

Meta-Generalized Gradient Approximation Made Magnetic

Jacques K. Desmarais,[1,*] Alessandro Erba,[1] Giovanni Vignale,[2] and Stefano Pittalis[3,†]
[1]*Dipartimento di Chimica, Università di Torino, via Giuria 5, 10125 Torino, Italy*
[2]*Institute for Functional Intelligent Materials, National University of Singapore, 4 Science Drive 2, Singapore 117544*
[3]*Istituto Nanoscienze, Consiglio Nazionale delle Ricerche, Via Campi 213A, I-41125 Modena, Italy*
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PHYSICAL REVIEW LETTERS **136**, 016403 (2026)

Physical Spin Torques from Exactly Constrained Exchange-Correlation Torques

Jacques K. Desmarais,[1,*] Kamel Bencheikh,[2] Giovanni Vignale,[3] and Stefano Pittalis[4,†]
[1]*Dipartimento di Chimica, Università di Torino, via Giuria 5, 10125 Torino, Italy*
[2]*Setif 1 University-Ferhat Abbas, Faculty of Sciences, Department of Physics* https://ror.org/02rzqza52 *y of Quantum Physics and Dynamical Systems, Setif, Algeria*
[3]*Institute for Functional Intelligent Materials, National University of Singapore, 4 Science Drive 2, Singapore 117544, Singapore*
[4]*Istituto Nanoscienze, Consiglio Nazionale delle Ricerche, Via Campi 213A, I-41125, Modena, Italy*
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- **Challenge 1:**
  Over-magnetization of MGGAs
- **Challenge 2:**
  Spurious xc-torques in Spin-DFT
- **Challenge 3:**
  Spin-orbit-coupling is not (always) “small”

In principle...

SpinCurrent-DFT

$$E = \min_{\Psi} \langle \Psi | \hat{H}_{\mathrm{SCDFT}} | \Psi \rangle$$

$$= \min_{(n,\, \vec{m},\, \mathbf{j},\, \vec{\mathbf{J}})} \Big\{ \min_{\Phi \to (n,\, \vec{m},\, \mathbf{j},\, \vec{\mathbf{J}})} \langle \Phi | \hat{T} | \Phi \rangle + E_{\mathrm{Hxc}}[n,\, \vec{m},\, \mathbf{j},\, \vec{\mathbf{J}}]$$

$$+ \int d^3r\, n\tilde{v} + \int d^3r\, m^a \tilde{B}^a + \int d^3r\, \mathbf{j} \cdot \mathbf{A} + \int d^3r\, \mathbf{J}^a \cdot \mathbf{A}^a \Big\}$$

$$\tilde{v} = v + \frac{1}{2} A_\nu A_\nu$$

$$\tilde{B}^a = B^a + A_\nu A_\nu^a$$

Spin-orbit coupling

Rasolt, Vignale Phys. Rev. Lett. 59, 2360 (1987)

Rasolt, Vignale Phys. Rev. B 37, 10685 (1988)

Bencheikh J. Phys. A: Math. Gen. 36, 11929 (2003)

Kohn-Sham system in SpinCurrent-DFT (1/3)

$$\Phi_s(\mathbf{x}_1,\ldots,\mathbf{x}_N) = \frac{1}{\sqrt{N!}}\,\det\begin{bmatrix}\phi_1(\mathbf{r}_1) & \phi_2(\mathbf{r}_1) & \ldots & \phi_N(\mathbf{r}_1)\\ \phi_1(\mathbf{r}_2) & \phi_2(\mathbf{r}_2) & \ldots & \phi_N(\mathbf{r}_2)\\ \vdots & \vdots & \vdots & \vdots\\ \phi_1(\mathbf{r}_N) & \phi_2(\mathbf{r}_N) & \ldots & \phi_N(\mathbf{r}_N)\end{bmatrix}$$

$$\phi(\mathbf{r}) = \begin{bmatrix}\phi_\uparrow(\mathbf{r})\\ \phi_\downarrow(\mathbf{r})\end{bmatrix}$$

yields the exact densities

$$n(\mathbf{r}) = \sum_{k=1}^{N}\phi_k^\dagger(\mathbf{r})\phi_k(\mathbf{r})$$

$$j_\nu = \frac{1}{2i}\sum_{k=1}^{N}\left\{\phi_k^\dagger\left[\partial_\nu\phi_k\right] - \left[\partial_\nu\phi_k^\dagger\right]\phi_k\right\}$$

$$\vec{m}(\mathbf{r}) = \sum_{k=1}^{N}\phi_k^\dagger(\mathbf{r})\vec{\sigma}\phi_k(\mathbf{r})$$

$$\vec{J}_\nu = \frac{1}{2i}\sum_{k=1}^{N}\left\{\phi_k^\dagger\vec{\sigma}\left[\partial_\nu\phi_k\right] - \left[\partial_\nu\phi_k^\dagger\right]\vec{\sigma}\phi_k\right\}$$

Kohn-Sham system in SpinCurrent-DFT (2/3)

$$\left\{\frac{1}{2}\left[-i\nabla+\boldsymbol{\mathcal{A}}\right]^2+\left(\mathcal{V}-\frac{1}{2}\vec{A}_\nu\cdot\vec{A}_\nu\right)+\hat{V}_{\rm Hxc}\right\}\phi_k=\epsilon_k\phi_k$$

$$\boldsymbol{\mathcal{A}}=\mathbf{A}+\vec{\mathbf{A}}\cdot\vec{\sigma}$$

$$\hat{V}_{\rm Hxc}\equiv\mathcal{V}_{\rm Hxc}-\frac{i}{2}\left[\partial_\nu\mathcal{A}_{{\rm xc},\nu}+\mathcal{A}_{{\rm xc},\nu}\partial_\nu\right]$$

$$\mathcal{V}_{\rm (Hxc)}=v_{\rm (Hxc)}+\vec{B}_{\rm (xc)}\cdot\vec{\sigma}$$

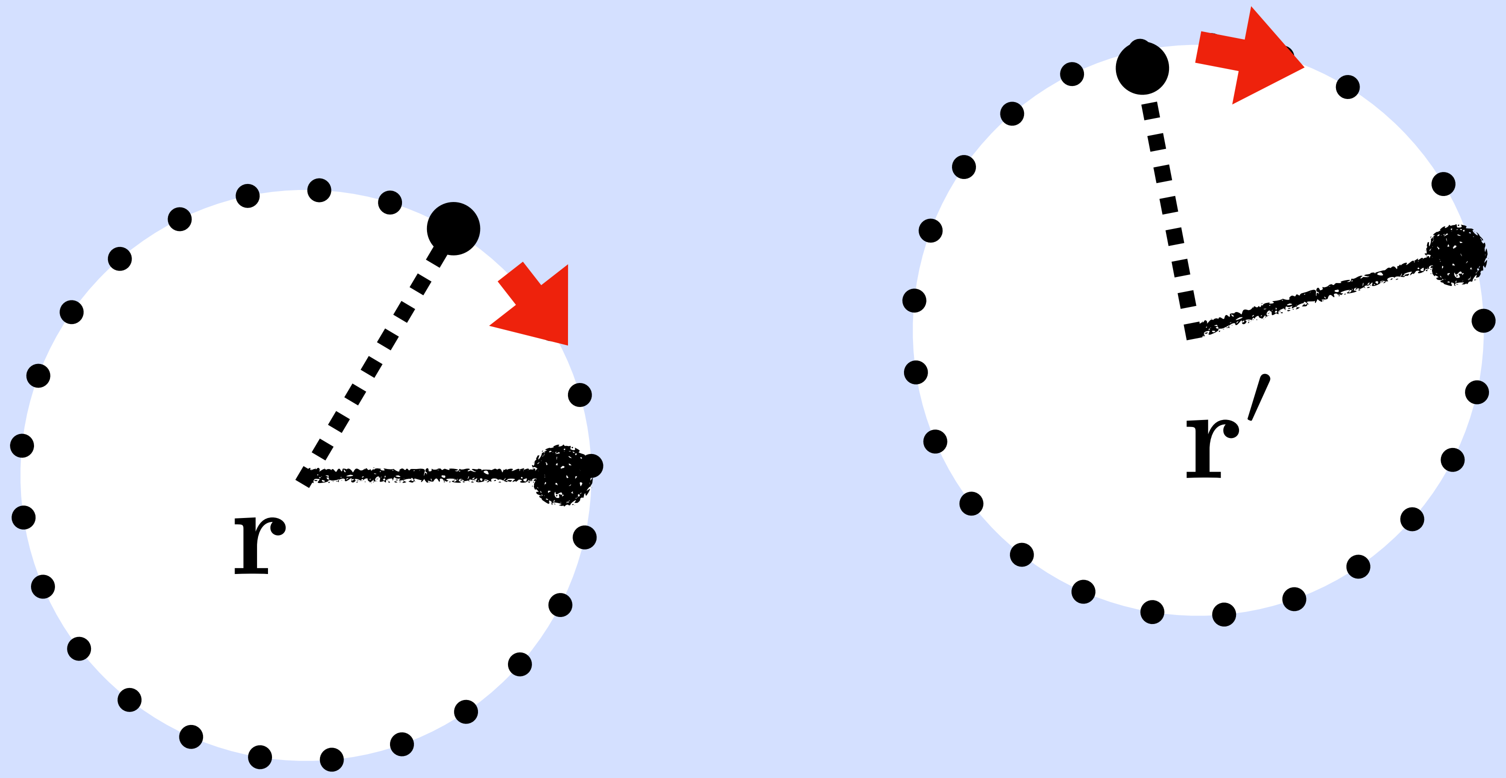
$$\mathcal{A}_{{\rm xc},\nu}=A_{{\rm xc},\nu}+\vec{A}_{{\rm xc},\nu}\cdot\vec{\sigma}$$

$$v_{\rm xc}=\frac{\delta E_{\rm xc}}{\delta n}\qquad \vec{B}_{\rm xc}=\frac{\delta E_{\rm xc}}{\delta\vec{m}}\qquad A_{{\rm xc},\nu}=\frac{\delta E_{\rm xc}}{\delta j_\nu}\qquad \vec{A}_{{\rm xc},\nu}=\frac{\delta E_{\rm xc}}{\delta\vec{J}_\nu}$$

Consider U(1) x SU(2) transformations

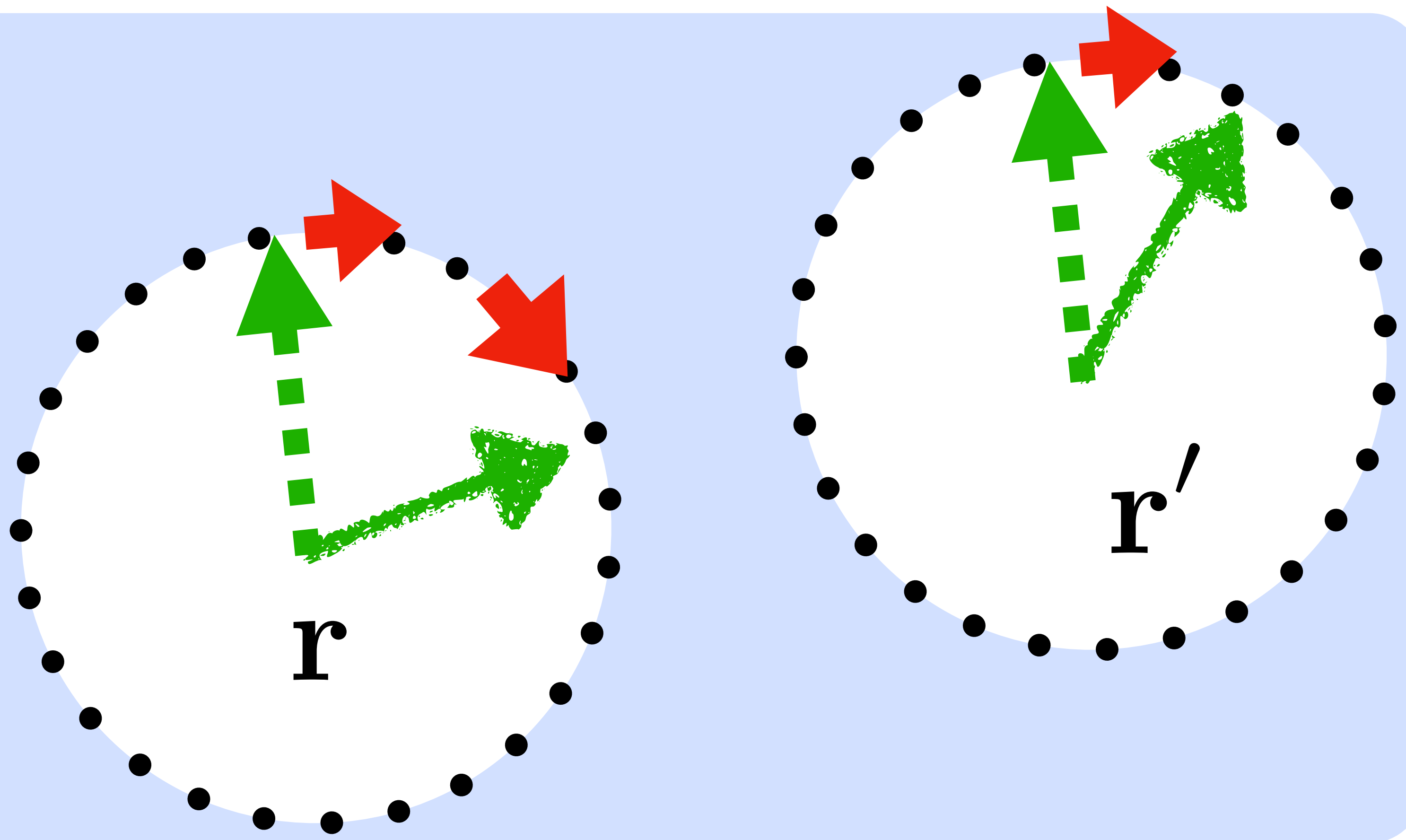
$$U(1): \quad \hat{\Psi}(\mathbf{r}) \to \hat{\Psi}(\mathbf{r}) \quad = \quad \exp\left[\frac{i}{c}\Lambda_0(\mathbf{r})\right]\hat{\Psi}(\mathbf{r})$$

Point-wise change of the orbital phases



$$SU(2): \quad \hat{\Psi}(\mathbf{r}) \to \hat{\Psi}'(\mathbf{r}) \quad = \quad \exp\left[\frac{i}{c}\vec{\Lambda}(\mathbf{r}) \circ \vec{\sigma}\right]\hat{\Psi}(\mathbf{r})$$

Point-wise change of spin orientations



Densities transform as follows

$$n \to n' = n$$

$$\mathbf{j} \to \mathbf{j}' = \mathbf{j} + \frac{1}{c} n \nabla \Lambda_0 + \frac{i}{2} \vec{m} \cdot \mathrm{Tr}\left(\vec{\sigma} U_\mathrm{S} \nabla U_\mathrm{S}^{-1}\right)$$

$$\vec{\mathbf{m}} \to \vec{\mathbf{m}}' = R(\vec{\Lambda}) \vec{\mathbf{m}}$$

$$\vec{\mathbf{J}} \to \vec{\mathbf{J}}' = R(\vec{\Lambda}) \left[\vec{\mathbf{J}} - \frac{1}{c} n \left(\nabla \Lambda_0\right) \vec{m} - \frac{i}{2} n \mathrm{Tr}\left(\vec{\sigma} U_\mathrm{S} \nabla U_\mathrm{S}^{-1}\right) \right]$$

Yet, the xc energy is invariant

As a result of
U(1)×SU(2) invariance of the e.e. interaction
and
equality of KS and real (interacting) densities:

$$E_{\mathrm{xc}}[n, \vec{m}, \mathbf{j}, \vec{\mathbf{J}}] = E_{\mathrm{xc}}[n, \vec{m}', \mathbf{j}', \vec{\mathbf{J}}']$$

Therefore:
we must approximate the xc-energy
via U(1)×SU(2) invariant quantities

Digging out the relevant building blocks

$$h_{\mathrm{x}}(\mathbf{r}, u) := \frac{1}{4\pi} \int d\Omega_{\mathbf{u}}\ h_{\mathrm{x}}(\mathbf{r}, \mathbf{r} + \mathbf{u})$$

$$h_{\mathrm{x}}(\mathbf{r}, u) = -\frac{n(\mathbf{r})}{2} \left(1 + \frac{\vec{m}(\mathbf{r}) \cdot \vec{m}(\mathbf{r})}{n^2(\mathbf{r})}\right) + C_{\mathrm{x}}^{\mathrm{nc}}(\mathbf{r}) u^2 + \dots$$

$$C_{\mathrm{x}}^{\mathrm{nc}} = \frac{1}{3} \Bigg\{ \left[\left(\tau - \frac{j_\nu j_\nu}{2n} \right) - \left(\frac{\nabla^2 n}{4} + \frac{(\partial_\mu n)(\partial_\mu n)}{8n} \right) \right] + \left[\left(\frac{\vec{m} \cdot \vec{\tau}}{n} - \frac{\vec{J}_\nu \cdot \vec{J}_\nu}{2n} \right) - \left(\frac{\vec{m} \cdot \nabla^2 \vec{m}}{4n} + \frac{(\partial_\mu \vec{m}) \cdot (\partial_\mu \vec{m})}{8n} \right) \right]$$

PRB 96, 035141 (2017)

Natural extra (non independent) densities

$$\tau = \frac{1}{2}\sum_{k=1}^{1}\left(\partial_\nu \phi_k^\dagger\right)\left(\partial_\nu \phi_k\right)$$

$$\vec{\tau} = \frac{1}{2}\sum_{k=1}^{1}\left(\partial_\nu \phi_k^\dagger\right)\vec{\sigma}\left(\partial_\nu \phi_k\right)$$

These densities can be handled via Generalized-KS of SpinCurrent-DFT

PRM 8, 013802 (2024)

From SCAN to nc-mSCAN

$$E_{\rm xc} = \int d^3r \; n(\mathbf{r})\epsilon_{\rm xc}^{\rm SCAN}(\mathbf{r})$$

$$\tau = \frac{1}{2}\sum_j^{\rm occ} |\nabla\varphi_j|^2$$

$$\epsilon_{\rm xc}^{\rm SCAN} = \epsilon_{\rm xc}^1 + \left(\epsilon_{\rm xc}^0 - \epsilon_{\rm xc}^1\right) f_{\rm xc}(\alpha)$$

$$\alpha = (\tau - \tau_{\rm W})/\tau_{\rm unif}$$

$${\rm ELF} = 1/(1+\alpha^2)$$

$$\tau \to \tilde{\tau} = \left(\tau - \frac{j_\nu j_\nu}{2n}\right) + \left(\frac{\vec{m}\cdot\vec{\tau}}{n} - \frac{\vec{J}_\nu \cdot \vec{J}_\nu}{2n}\right) + \frac{\partial_\nu \vec{m}\cdot\partial_\nu \vec{m}}{8n}$$

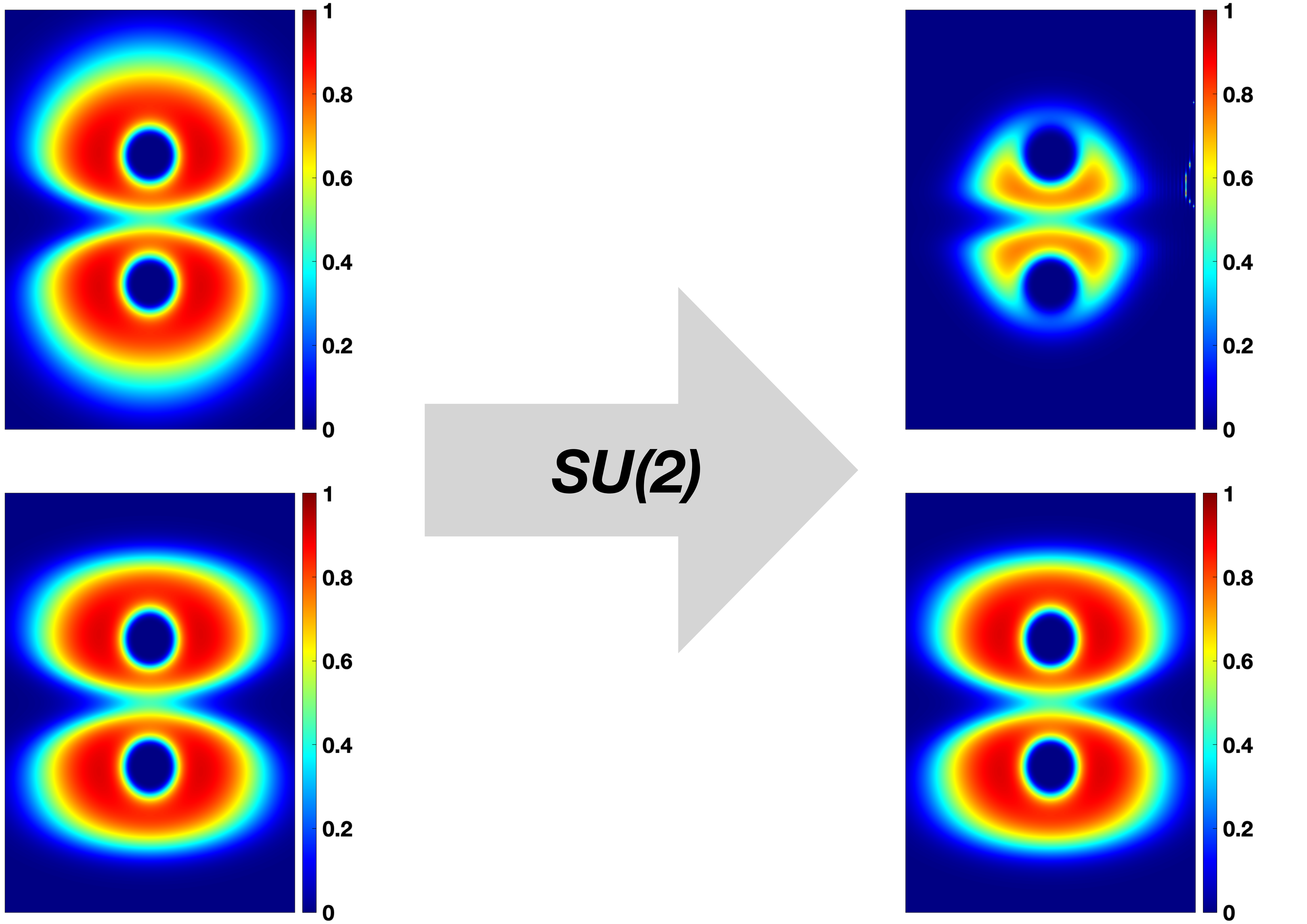
Note: the LDA "limit" of SCAN should also be replaced with the nc-LSDA (not shown here).

PRL 134, 106402 (2025)

Gauge-invariant Non-collinear ELF

Näive extension applied to $\mathbf{I}_2^-$

Proper extension applied to $\mathbf{I}_2^-$



PRL 133, 136401 (2024)

Generalized-Kohn-Sham equation (3/3)

$$\left\{\frac{1}{2}\left[-i\nabla + \boldsymbol{\mathcal{A}}\right]^2 + \left(\mathcal{V} - \frac{1}{2}\vec{A}_\nu \cdot \vec{A}_\nu\right) + \hat{V}^{\mathrm{MGGA}}_{\mathrm{Hxc}}\right\}\phi_k = \epsilon_k \phi_k$$

$$\hat{V}^{\mathrm{MGGA}}_{\mathrm{Hxc}} \equiv \mathcal{V}_{\mathrm{Hxc}} - \frac{1}{2}\partial_\nu \mathcal{M}_{\mathrm{xc}} \partial_\nu - \frac{i}{2}\left[\partial_\nu \mathcal{A}_{\mathrm{xc},\nu} + \mathcal{A}_{\mathrm{xc},\nu}\partial_\nu\right]$$

$$\mathcal{M}_{\mathrm{xc}} = M_{\mathrm{xc}} + \vec{M}_{\mathrm{xc}} \cdot \vec{\sigma}$$

$$M_{\mathrm{xc}} = \frac{\partial \epsilon_{\mathrm{xc}}}{\partial \tau} \quad \vec{M}_{\mathrm{xc}} = \frac{\partial \epsilon_{\mathrm{xc}}}{\partial \vec{\tau}}$$

PRM 8, 013802 (2024)

PRL 136, 016403 (2026)

Does it work?

Challenge 1: Over-magnetization of MGGAs

Eur. Phys. J. B (2018) 91: 193
https://doi.org/10.1140/epjb/e2018-90275-5
THE EUROPEAN PHYSICAL JOURNAL B
Regular Article

From one to three, expl in magnetic alloys*

Aldo H. Romero[1,a] and Matthieu J. Verstraete

PHYSICAL REVIEW LETTERS **121**, 207201 (2018)
Editors' Suggestion

Applicability of the Strongly Constrained and Appropriately Normed Density Functional to Transition-Metal Magnetism

Yuhao Fu and David J. Singh*
Department of Physics and Astronomy, University of Missouri, Columbia, Missouri 65211-7010, USA

LDAs and GGAs can be better than MGGAs

PHYSICAL REVIEW B **100**, 041113(R) (2019)
Rapid Communications

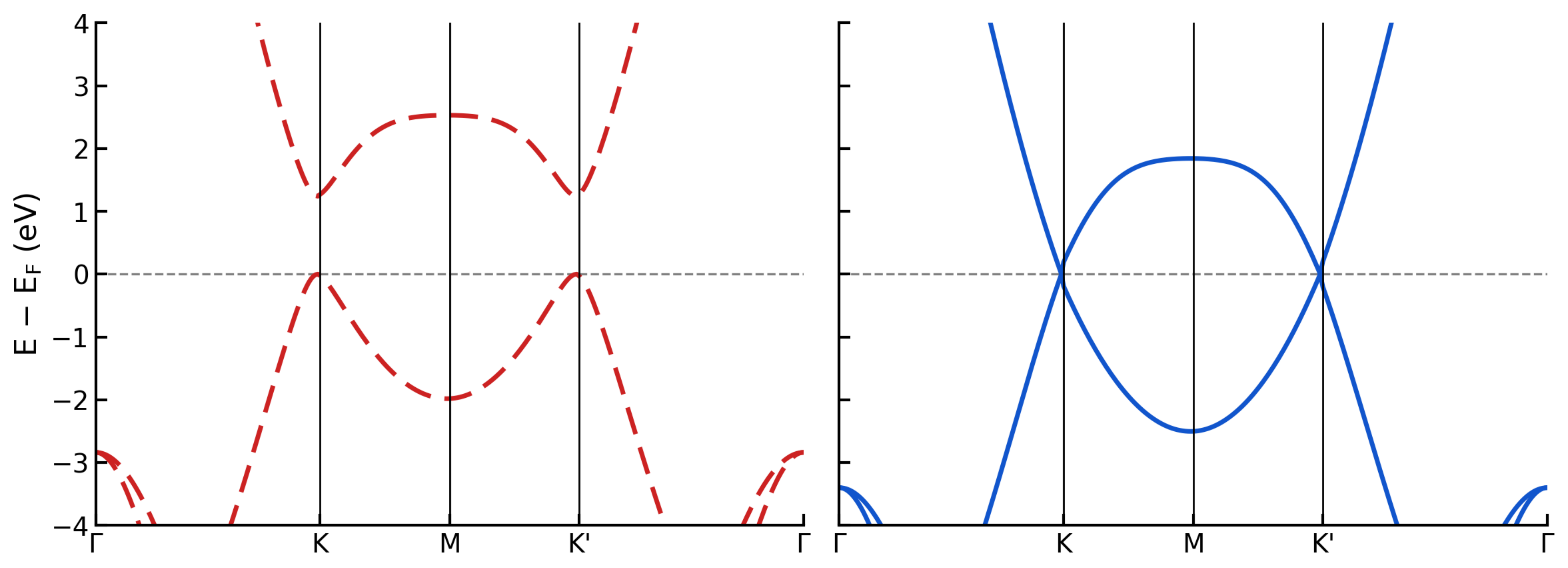
Analysis of over-magnetization of elemental transition metal solids from the SCAN density functional

Daniel Mejía-Rodríguez* and S. B. Trickey†
Quantum Theory Project, Department of Physics, University of Florida, Gainesville, Florida 32611, USA

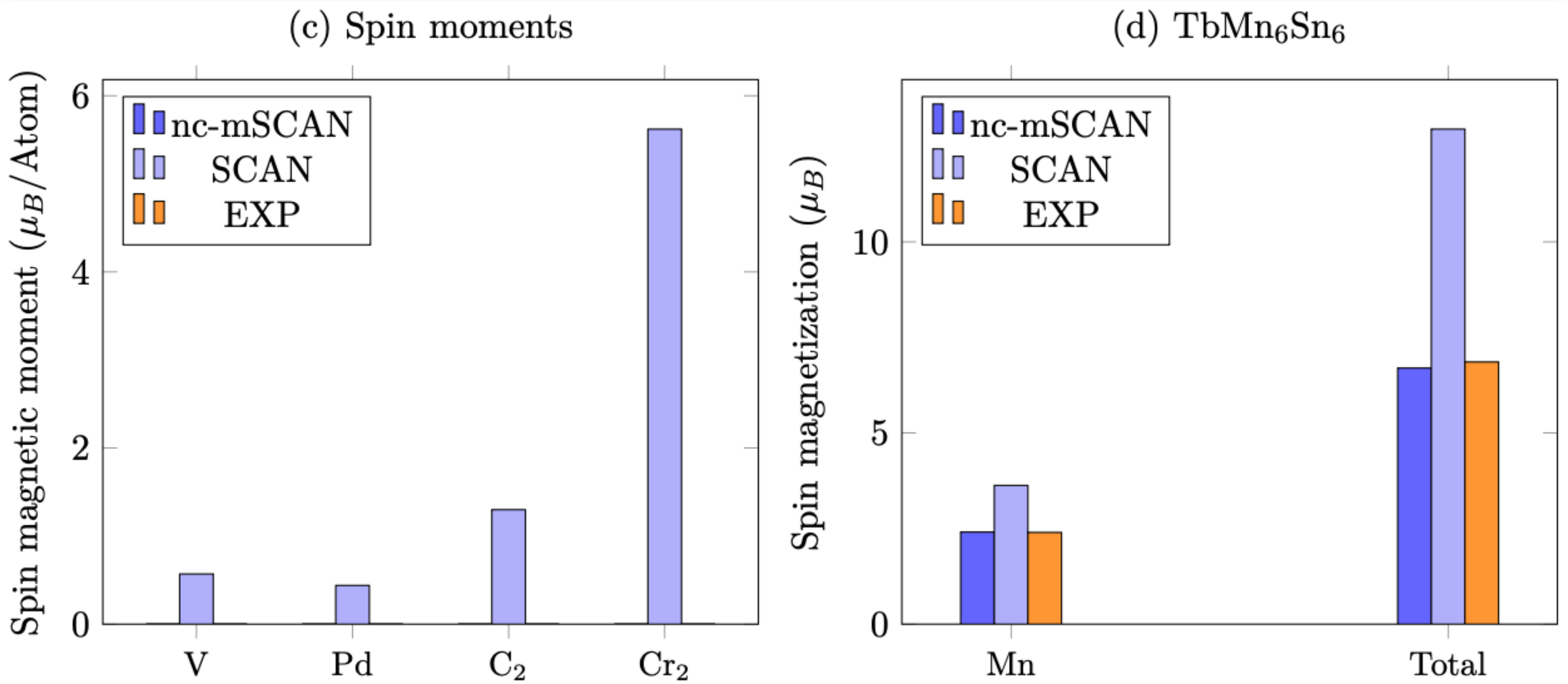
407 (2020)

Short describing magnetism

Fabien Tran,[1] Guillaume Baudes Madsen,[1] Peter Blaha,[1] Karlheinz Schwarz,[1]
[1]*Institute of Materials Chemistry, Vienn Getreidemarkt 9/165-TC, A-1060 Vienna, Austria*
[2]*Univ Rennes, ENSCR, CNRS, ISCR (Institut es de Rennes) - UMR 6226, F-35000 Rennes, France*
[3]*Department of Physics and Astronomy, Un ssouri, Columbia, Missouri 65211-7010, USA*



Graphene

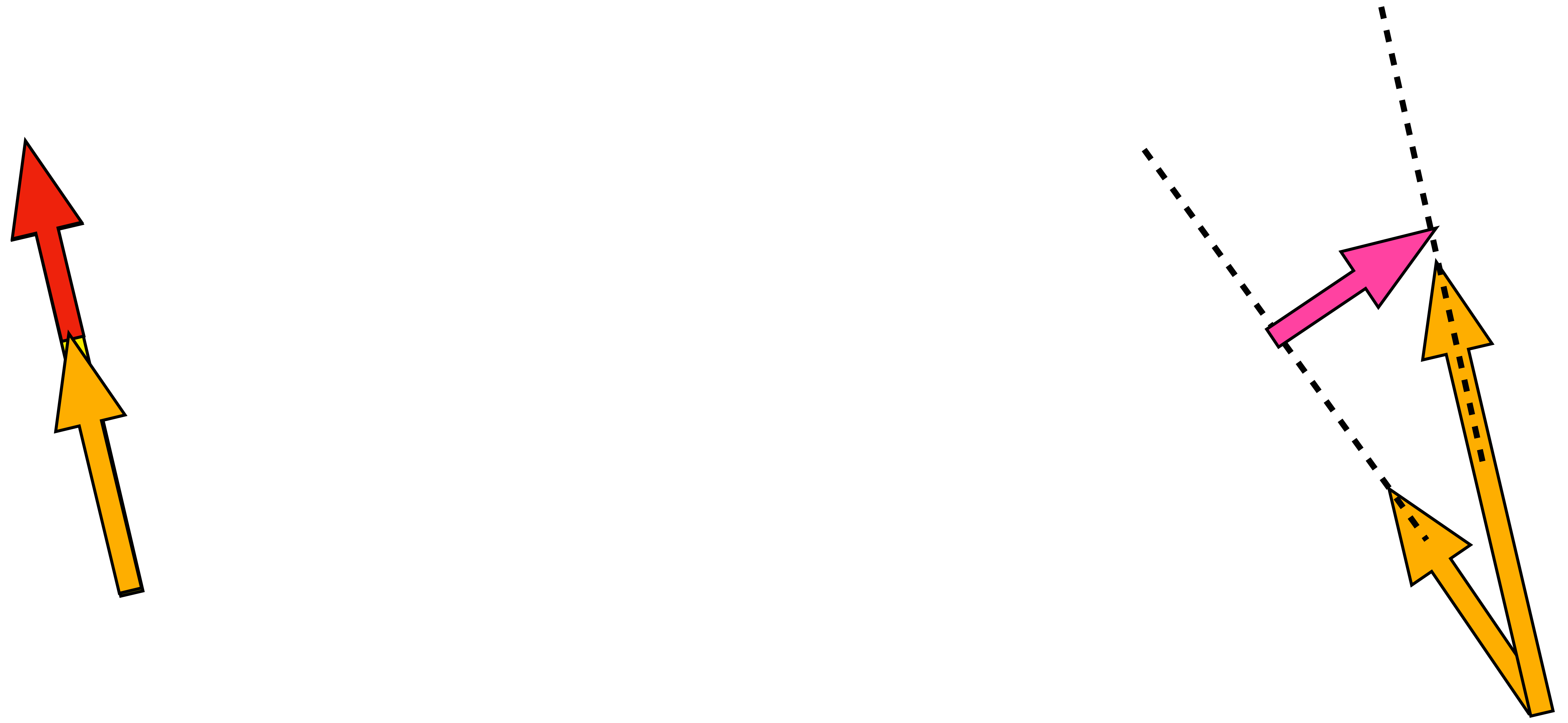


Spin moments (μ_B/Atom)

	Fe	Co	Ni	FeO	CoO	NiO
SCAN	2.62	1.74	0.66	3.67	2.65	1.62
nc-mSCAN	2.15	1.58	0.56	3.57	2.52	1.44
EXP	1.98-2.08	1.52-1.62	0.52-0.55	2.32-4.00	1.75-2.98	1.45-1.90

PRL 134, 106402 (2025)

Challenge 2: Spurious xc-torques



Longitudinal and Transverse spin gradients

Spin (paramagnetic) current

KS Spin current

$$\boxed{\vec{m} \times \vec{B}_{\mathrm{xc}}} = \partial_\nu \left[\vec{J}_\nu - \vec{J}_{s,\nu} \right]$$

Exact condition in SDFT [Capelle *et al.* PRL 87 (2001)]

Exact condition
but of marginal usefulness
for improving
approximations!

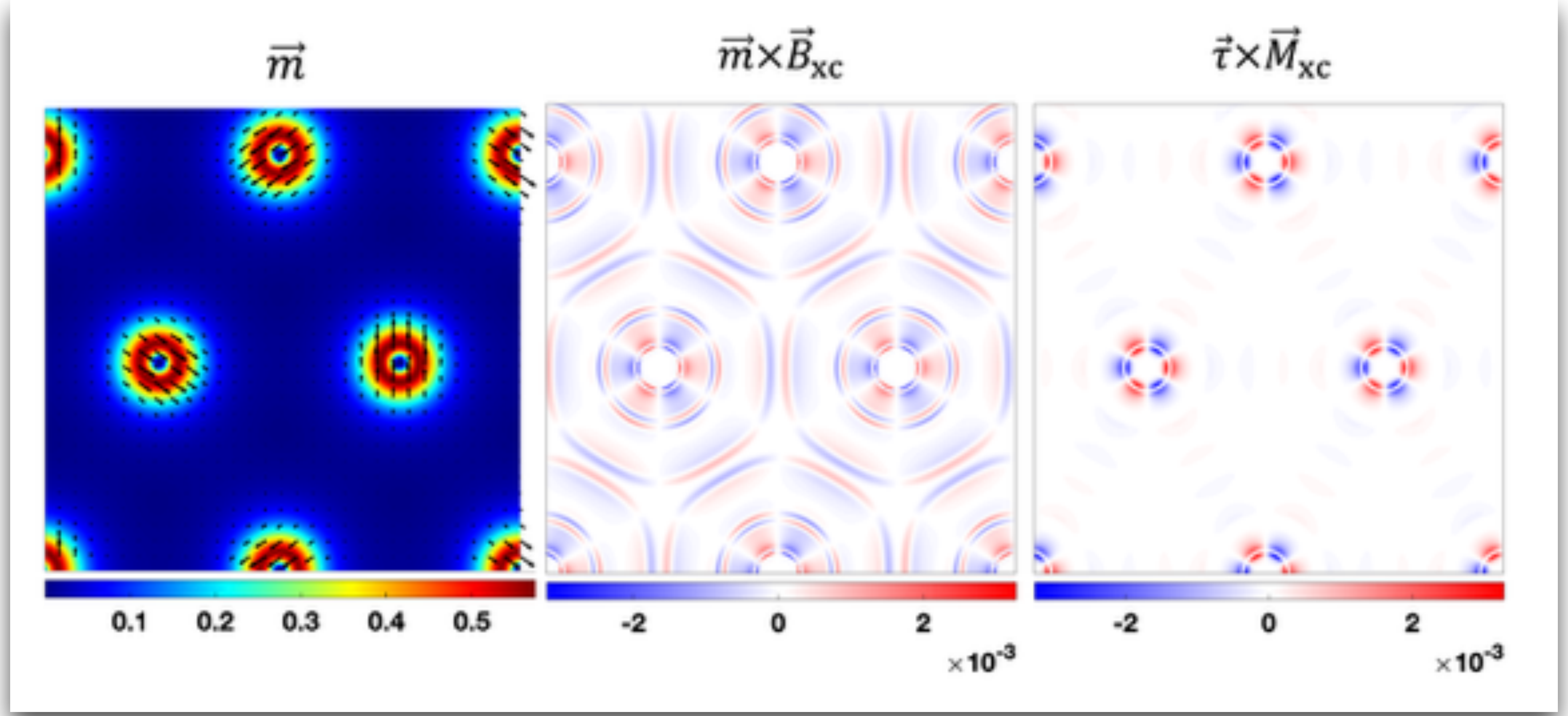
In addition:

An important source of spin currents is Spin-Orbit Coupling (SOC)

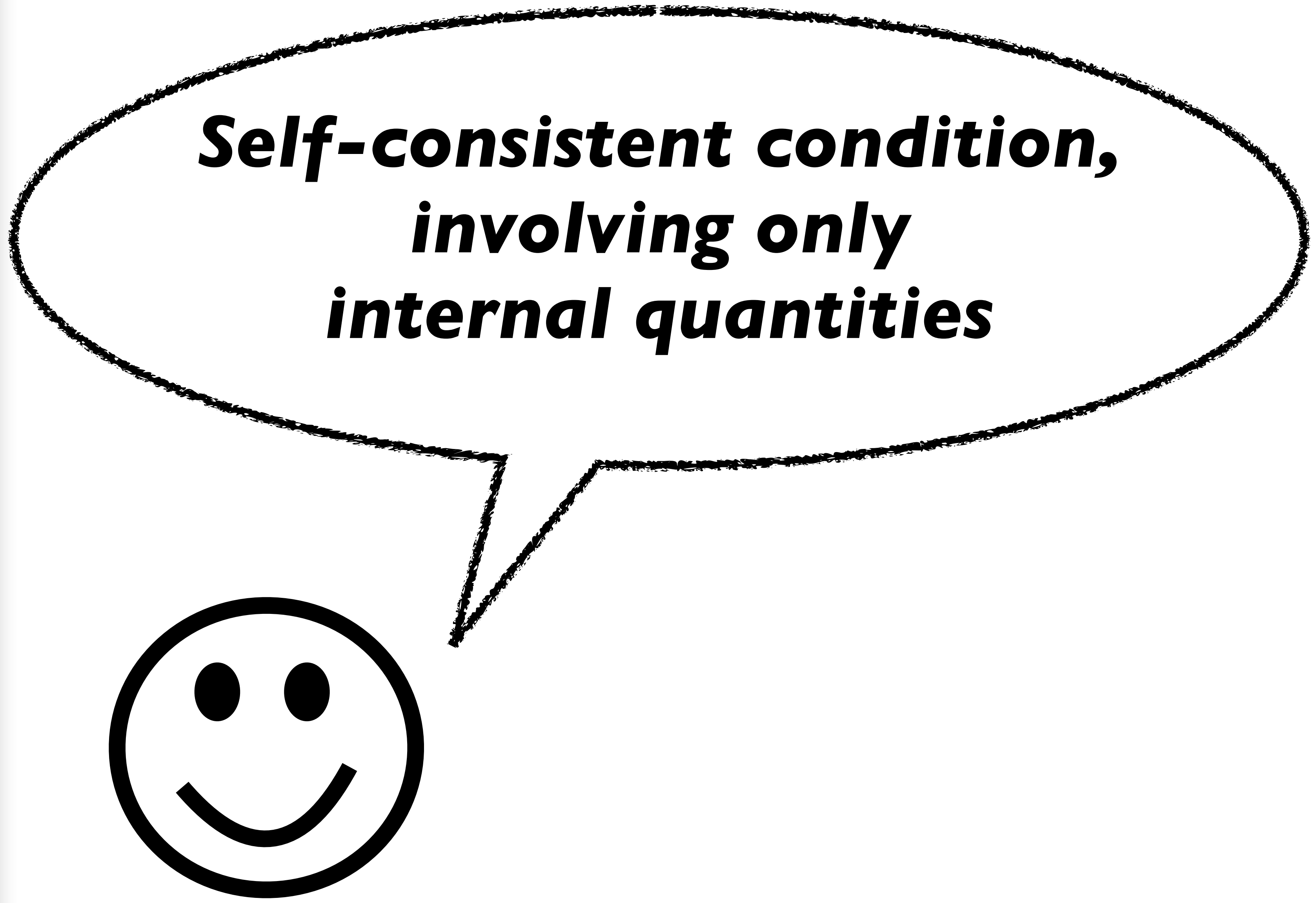
XC-torques of U(1)×SU(2)-MGGAs

$$\vec{m} \times \vec{B}_{\mathrm{xc}} = \frac{1}{4}\partial_\nu \left[(\partial_\nu \vec{m}) \times \vec{M}_{\mathrm{xc}} \right] - \vec{\tau}_{\mathrm{nc}} \times \vec{M}_{\mathrm{xc}}$$

Only internal (GKS) quantities!



Chromium (001) unsupported monolayer



PRL 136, 016403 (2026)

Challenge 2: Spurious xc-torques

$$\frac{\partial \vec{m}}{\partial t} + \partial_\nu \vec{J}_{s,\nu} = 2\left(\vec{B} + \underline{\vec{B}_{\mathrm{xc}}}\right) \times \vec{m}$$

Time-Dependent Spin-DFT

Spurious

Different form from physical equation

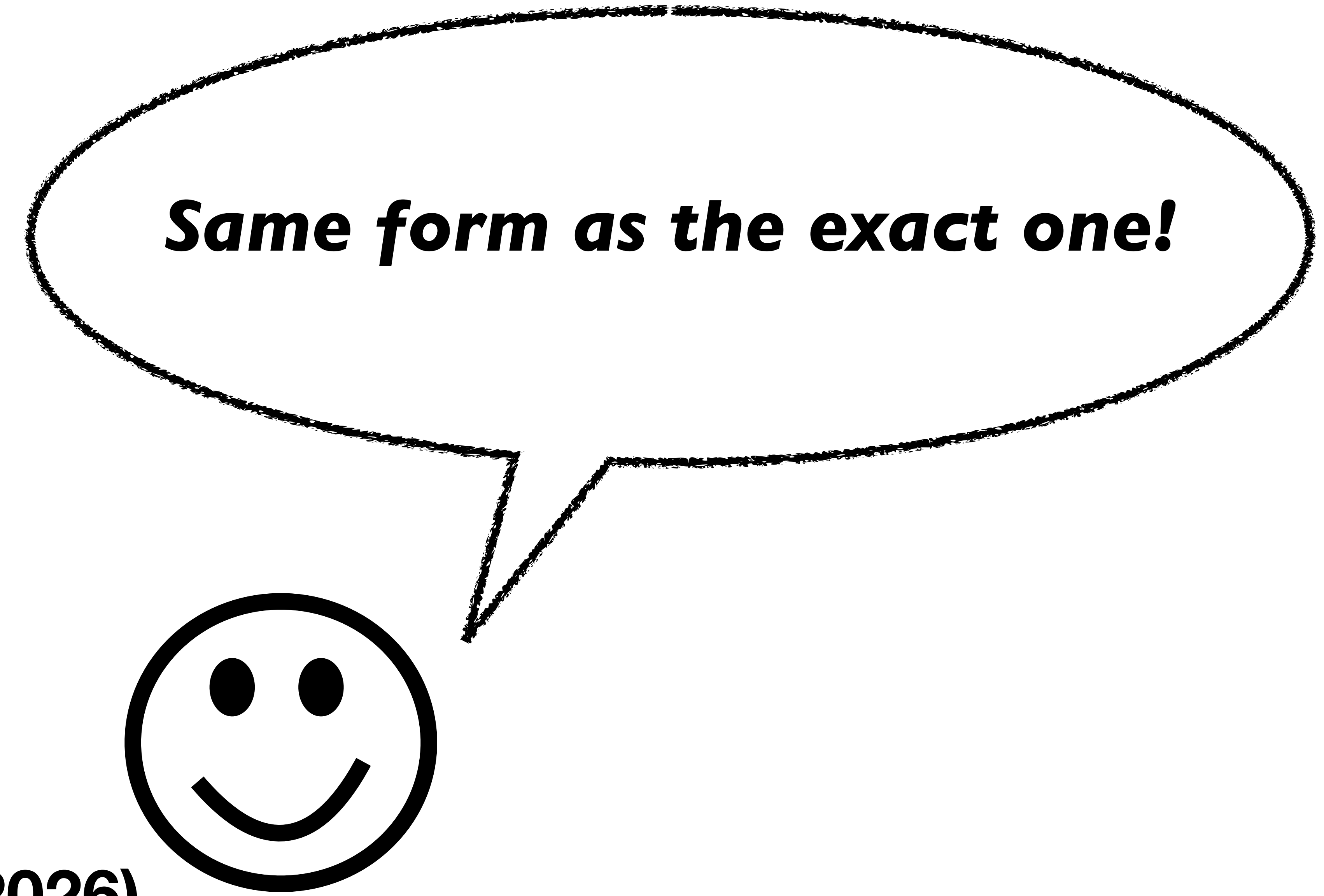
In addition:

An important source of spin currents is Spin-Orbit Coupling (SOC)

Adiabatic dynamics from U(1)×SU(2)-MGGAs

$$\frac{\partial n}{\partial t} + \partial_\nu \left[j_\nu + \frac{1}{c} n A_\nu + \frac{1}{4} \vec{m} \cdot \vec{A}_\nu \right] = 0$$

$$\frac{\partial \vec{m}}{\partial t} + \partial_\nu \left[\vec{j}_\nu + \vec{m} A_\nu + \frac{1}{4} n \vec{A}_\nu \right] = \vec{B} \times \vec{m} + \frac{1}{2} \vec{A}_\nu \times \vec{J}_\nu$$

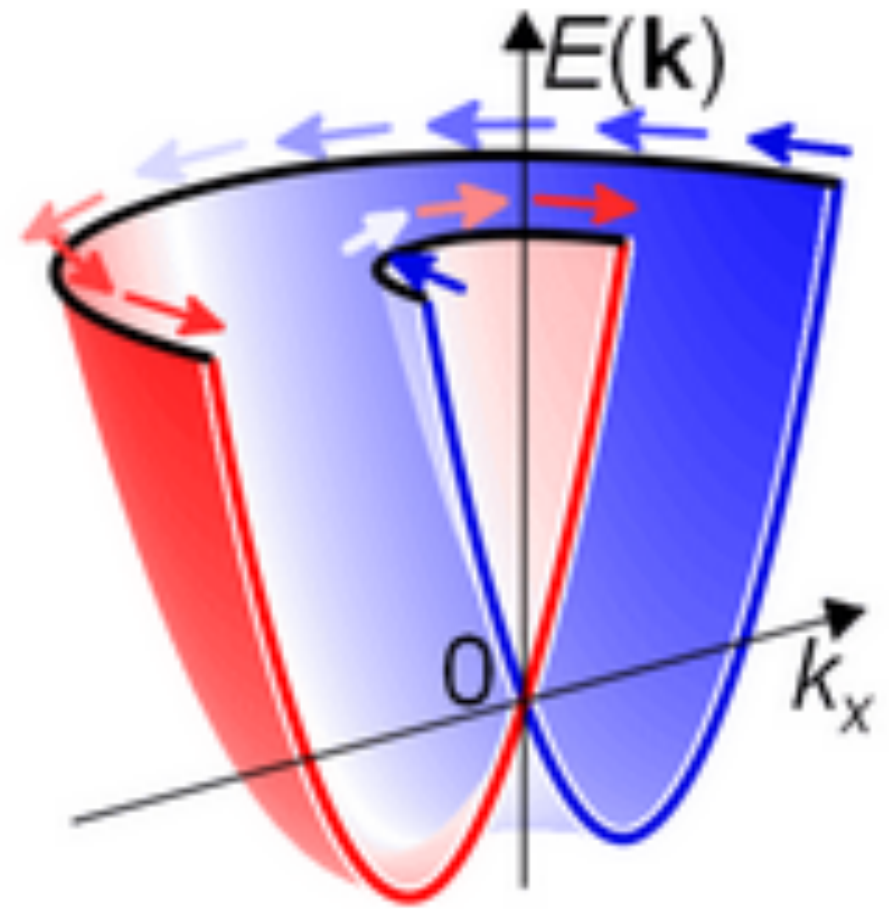


$$\mathcal{M}_{\mathrm{xc}} = M_{\mathrm{xc}} + \vec{M}_{\mathrm{xc}} \cdot \vec{\sigma}$$

$$M_{\mathrm{xc}} = \frac{\partial \epsilon_{\mathrm{xc}}}{\partial \tau} \quad \vec{M}_{\mathrm{xc}} = \frac{\partial \epsilon_{\mathrm{xc}}}{\partial \vec{\tau}}$$

PRL 136, 016403 (2026)

Problem 3: SOC is not (always) “small”



Rashba-type spin split band structure

$$\vec{m} = \vec{0}, \quad \mathbf{j} = \mathbf{0}$$

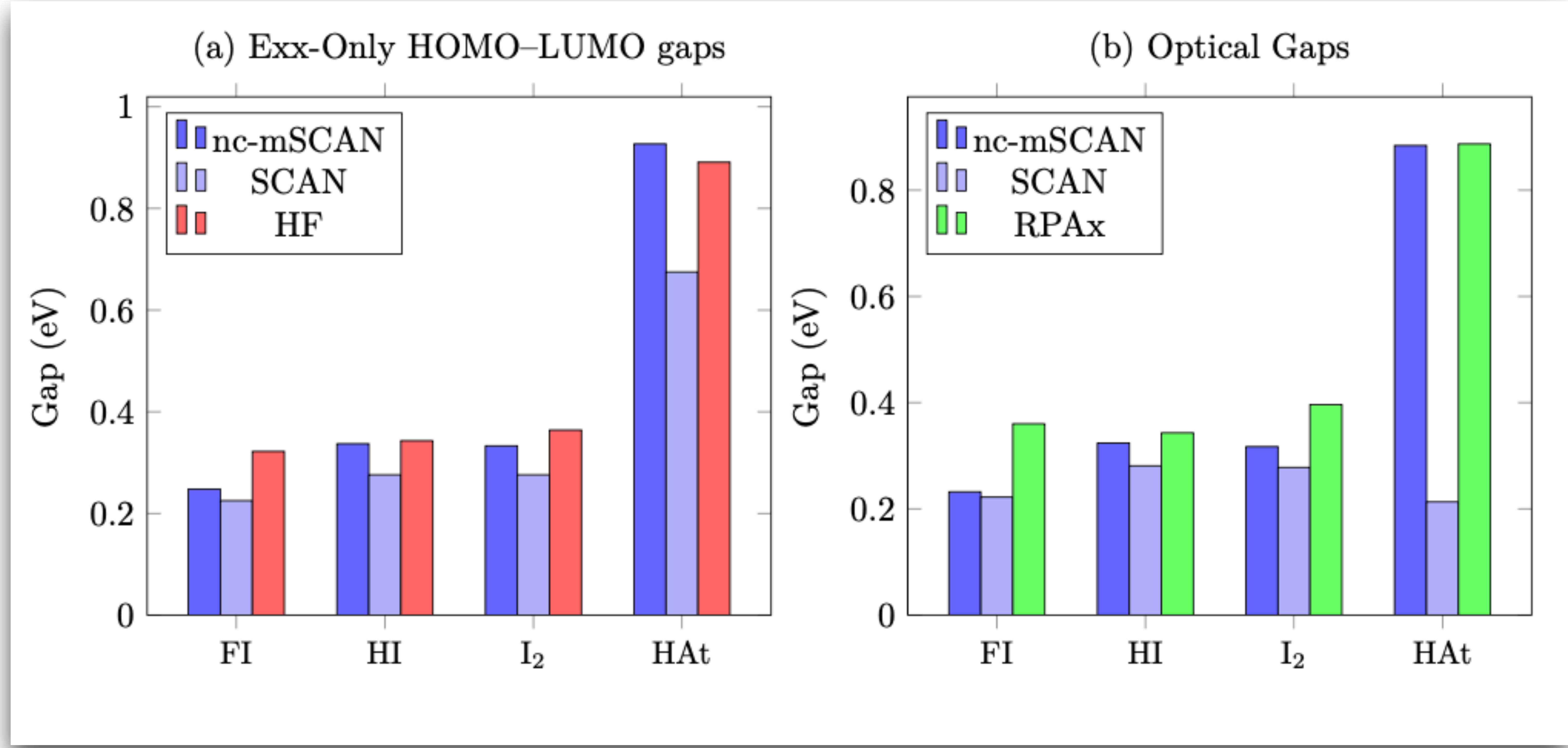
$$\vec{\mathbf{J}} \neq \vec{\mathbf{0}}$$

Usually treated non-self-consistently!

No feedback from spin currents.

Splitting of valence band at $\mathbf{K}$

	2D $MoSe_2$ [eV]	3D α-$MoTe_2$ [eV]	2D WSe_2 [eV]
r^2 SCAN	0.14	0.28	0.43
J-r^2 SCAN	0.20	0.32	0.52
Exp.	0.18	0.30–0.34	0.52



Full self-consistency!
Spin currents included

PRL 132, 256401 (2024)

Conclusion: Does it work?

Yes, it does:
Exact conditions do <u>M</u>atter!